
Context Window Controller

Evidence review summary and implementation stance

netg

June 2026 vv0.1 first pass

Executive verdict

15

FINDINGS SURVIVED

0

FINDINGS DISCARDED

1

BLOCKING UNCERTAINTY

Bottom line: the CWC architecture is sound enough to implement.

Guardrail: arrival demotion must own `persist → verify → replace`. Do not assume the runtime already retained byte-exact tool output.

What the findings validate

Cache economics - Prefix caches are sensitive to the first changed token - Tail appends are cheap - Middle rewrites must be batched - Anthropic cache fields make this measurable

Data safety - Manifest stores pointers, not payloads - Hindsight recall is not byte-exact retrieval - Sidecar remains prudent until handles are proven - Summaries must replay byte-identically

Design implication: reduce live prefix size early, but only through cache-safe surfaces.

The one unresolved premise

Unproven: every tool output is synchronously retained with a usable handle before arrival-plane rewriting.

Required implementation contract

1. Capture raw tool output
2. Retain raw payload to Hindsight or sidecar
3. Verify the handle resolves byte-exactly
4. Only then emit the envelope or stub
5. If any step fails, keep the raw output live

```
raw output → retain → verify → replace  
no verify → no replace
```

Runtime risks to probe first

AREA	RISK	REQUIRED BEHAVIOR
Claude Code hooks	<code>updatedToolOutput</code> may be ignored for some tools/versions	Version-gate and fail closed
Tool failures	Failure path may not honor replacement	Test <code>PostToolUseFailure</code> behavior
Hindsight wrapper	Existing wrapper may only expose <code>retain / recall</code>	Add get-by-handle or sidecar
LiteLLM	Proxy layer does not own canonical history	Do not use as curation layer

What CWC should store

Manifest: hot index only

- Span identity and session ownership
- Sequence position and span type
- Hindsight ref and optional blob hash
- Raw/live token counts
- State: live or demoted
- Frozen summary envelope
- Safety flags: pinned, load-bearing, recent, active edit
- Access/demotion metadata for tuning

Never store raw tool or file payloads in the manifest.

The safe operating model



Implementation stance

PHASE	BUILD	SHIP RULE
1	Manifest + arrival plane	Fail closed to raw output
2	Batched curation plane	One thresholded cache break only
3	Retrieve / recall + tuning	Tune toward low, nonzero refetch rate

Start narrow: Phase 1 gives immediate token savings with minimal cache risk.

Safety rules that become tests

- No per-turn middle edits
- Stable zone is immutable except scheduled summary-anchor updates
- Recent-K, pinned, load-bearing, open-task, and active-edit spans are never demoted
- Demoted summaries are generated once and replayed byte-for-byte
- Missing handle or blob means no demotion
- Re-fetched content returns at the tail

Next actions

1

PROBE HOOK MATRIX

2

VERIFY BYTE-EXACT RETRIEVAL

3

IMPLEMENT PHASE 1

Decision

Proceed with CWC, but treat arrival retention as CWC-owned until the target runtime proves otherwise.